

Medha Rakesh

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EDUCATION

University of California, Berkeley

Berkeley, CA

Bachelor of Arts in Data Science, Computer Science

May 2028

- **Relevant Coursework:** Foundations of Data Science, Structure and Interpretation of Computer Programs, Data Structures, Linear Algebra & Differential Equations, Principles and Techniques of Data Science, Discrete Math & Probability

EXPERIENCE

Bloomberg

July 2026

Data Science Instructor, Kode With Klossy

San Francisco, CA

- Guided **30 high school scholars** through the full **software development lifecycle** in a two-week intensive, from requirements and wireframes to code review and final demo presentations, while teaching SQL and relational database design

Legali AI

September 2026 – Present

Software Engineering Intern | Next.js, React, TypeScript, Framer Motion, Tailwind

Remote

- Shipped a **full redesign** of the company's public homepage for Lea, an agentic AI legal platform expanding **access to justice** for survivors and self-represented users, building it end to end in Next.js, React, TypeScript, and Framer Motion
- Improved site usability and accessibility by building **keyboard-accessible components**, persistent light/dark theming, and motion-driven sections, and restructured the site's **information architecture** to consolidate user conversion flows
- Removed the legacy pricing path and validated the redesign through linting and a **clean production build** before review

GitMachine

September 2026 – Present

Machine Learning Engineer Intern | Python, PyTorch, vLLM, CUDA, MXFP8

Remote

- Built an **end-to-end onboarding process** for open-source LLM model families on the company's **GPU serving stack**, covering checkpoint audits, serving-engine verification, and pre-deployment memory and latency prediction for each model
- Verified each onboarded model serves correctly and meets **performance targets** through validation experiments, and audited an **MXFP8**-quantized checkpoint using the vendor's recipe over framework defaults, documenting its limits

HP

September 2026 – Present

ML Systems Researcher (Contract) | Python, PyTorch, llama.cpp, vLLM, Docker

Berkeley, CA

- Built a **reproducible benchmark harness** capturing **14 metrics** per run (TTFT, tokens/sec, memory, power) with **one-command reruns** by containerizing llama.cpp, Ollama, PyTorch, and vLLM with pinned versions in **Docker**
- Benchmarked LLM inference on the HP ZGX Nano across **5 model size tiers (7B–200B)** 3 quantization levels, and 128K contexts, and built a cost model comparing dollars per million tokens against cloud GPU rental and frontier APIs

Synopsys

December 2025 – May 2026

Software & ML Systems Engineer (Contract) | C/C++, Django, Redis, Jest, Git

Remote

- Surfaced real-time power and simulation metrics across **50+** hardware benchmarks for Synopsys chip design engineers by engineering a C++ and Django visualization dashboard with parameterized **REST API** endpoints and **WebSocket** feeds
- Cut repeat API latency by **68%** across **10,000+** chip design simulations by layering **Redis** caching onto hot routes
- Saved **2+ simulation hours** per sign-off by training **gradient-boosted models** to forecast chip power from netlist features

The SALT Research Group

January – May 2026

Machine Learning Research Assistant

Berkeley, CA

- Replaced a **fully manual workflow** by engineering an **automated data processing pipeline** that parses raw instrument output files, computes summary statistics across experimental trials, and exports structured data for downstream analysis
- Measured the electrical conductivity of FLiBe molten salt across **500–700°C** for **nuclear reactor safety monitoring** by implementing EIS signal preprocessing logic, including frequency filtering, sign-convention correction, and trial averaging

PROJECTS

ContractAlpha | Python, FastAPI, Next.js, TypeScript

Top 8 Finalist of ~300 teams

- Built a full-stack platform turning USAspending.gov federal contract award data into stock signals; engineered a date-cutoff layer preventing **lookahead bias** and parallelized batched LLM calls, outperforming the **S&P 500 by 24.7%** in backtests

Mental Galaxy | FastAPI, WebSockets, Deepgram, Claude, Sentry

UC Berkeley AI Hackathon 2026

- Built the **real-time** FastAPI + WebSocket backend streaming live mic audio through Deepgram speech-to-text and Claude classification; fixed a **production WebSocket crash** and led integration of three developers' components into one system

Cloudscape | Next.js, AWS SDK, Perplexity Sonar, Supabase

Best Use of Perplexity Agent API

- Built an AI incident commander that pulls CloudWatch metrics, logs, and CloudTrail events **in parallel** when an alarm fires, cutting **20–30 minutes** of manual triage and returning a likely root cause, ranked fix commands, and a full postmortem

SKILLS

Languages & Frameworks: Python, Java, C++, TypeScript, JavaScript, SQL, HTML/CSS, React, Next.js, Node.js, FastAPI, Django, Framer Motion

Infra & Tools: Docker, Kubernetes, AWS, Redis, Supabase, REST APIs, WebSockets, Git, Sentry, PyTorch, vLLM, llama.cpp